崔京伟

食品科学博士（在读）| 感官与消费者科学 × 感知数据科学 | 风味研究

核心优势

感官与消费者研究：系统掌握 QDA 定量描述分析、三点检验、偏好测试等感官方法学与实验设计；具备评价小组的组建、筛选、培训与一致性校准经验；主导多项企业委托的感官评价与风味筛选项目，熟悉人类受试研究的数据采集规范与质量控制；并完成个体感知差异的方差分解与消费者偏好分群建模。

感知数据科学与 AI 建模：精通 Python、MATLAB 与 SQL，在感知研究场景中独立完成神经网络、集成学习、随机森林、XGBoost、高斯过程回归等模型的构建、调优与验证；熟练运用方差分析、主成分分析等多元统计方法，将高维感知数据转化为可解释、可决策的研究结论。

多源数据整合与端到端自动化：将仪器传感数据（电子鼻、电子舌、GC-MS）、人类感官评价数据与消费者偏好数据在同一建模框架下融合；独立开发 GC-MS 数据自动化处理系统（累计处理样品 3000+）与电子鼻数据自动提取代码（累计处理样品 5000+），实现从原始输出到建模输入的全流程自动化；并主导规范化分析数据库的设计与数据治理。

风味化学与仪器分析：独立完成 GC-MS / GC-O / GC×GC-MS / 电子鼻 / 电子舌 全套仪器分析，熟悉关键香气化合物的定性定量与 OAV / ROAV 表征，能够将仪器结果与感官结论有效对接。

科研产出与学术表达：第一作者 SCI 论文 8 篇（5 篇已发表 / 3 篇审稿中），发明专利 2 项（均已授权），IUFoST 2026 世界大会报告 2 项（已录用）；具备面向技术与非技术受众、以及向管理层汇报研究结论的英文表达能力。

教育背景

2023.09 – 2027.06（预计） 江南大学 食品科学与工程（油脂及植物蛋白） 博士

导师：王兴国 教授

└ 2025.09 – 2027.06 新加坡国立大学 食品科学与技术 联合培养（导师：刘绍泉 教授）

2020.09 – 2023.06 江南大学 食品科学与工程（油脂及植物蛋白） 硕士

导师：金青哲 教授 | 学位论文盲审双优

2016.09 – 2020.06 东北农业大学 食品科学与工程（食品科学与安全） 学士

技术开发

风味 AI 工具集（个人网站） | 独立开发者 | redamancy564.github.io

独立设计与开发 8 个面向风味与感知研究的浏览器端 AI 工具，覆盖从分子、配方、仪器数据解析到消费者感知差异的完整链路。

核心功能：由分子结构预测气味与味道描述符；5000 个香味分子的气味空间可视化地图；按风味需求检索原料与替代分子、个体差异与消费者分群；GC-MS 峰表自动解析并输出关键香气化合物 OAV / ROAV 排名。

全部计算在浏览器本地完成，原始数据不上传服务器，兼顾易用性与数据保密要求。

GC-MS 数据自动化处理系统 | **独立开发者**

独立完成全流程软件方案的设计、开发与维护，实现 GC-MS 手动积分后数据的自动提取与深度分析。

核心功能：保留时间自动匹配、相邻峰智能聚合、单峰多候选物时基于规则的风味化合物精准判定。

落地效果：将原本数小时的二次数据处理压缩至分钟级，累计处理样品 3000+，并彻底消除人工转录误差，已用于科研数据的常规处理。

电子鼻数据自动化处理代码 | **独立开发**

从原始 txt 文件自动提取数据，完成自动基线校准与电阻值计算，替代人工逐条处理。

累计处理电子鼻样品 5000+，为后续建模提供标准化数据输入。

企业合作研发项目

某餐饮连锁企业 | **感官评价负责人** 2024.11 – 2024.12

风味筛选与感官评价

主导感官评价方案的设计与执行，从多款原型样品中完成系统的风味筛选。

结合仪器风味剖析与感官评价交叉验证，建立客观、可复现的评价流程。

某连锁饮品企业 | **项目负责人** 2025.01 – 2025.07

咖啡豆香气剖析与风味评价体系建立

应用 GC-O-MS 与 ROAV 表征咖啡豆的香气特征，建立系统的风味剖析与评价方法。

搭建可执行的风味筛选与评价流程，支撑产品风味研发。

某外资乳品/婴幼儿营养企业 | **核心成员** 2023.09 – 2024.12

乳成分分析与脂质数据库构建

建立标准化前处理与 GC 定量分析流程，完成大批量乳样本脂肪酸的高通量精准检测。

主导脂质数据库的设计与数据整理，形成规范、可复用的分析数据管理体系。

某方便食品企业 | **核心成员** 2023.12 – 2024.04

煎炸油质量控制与氧化稳定性分析

承担油脂质量控制与氧化稳定性评价，完成酸价、过氧化值、氧化稳定性指数等多指标测定。

构建油脂氧化降解的动力学分析方法，为货架期评估提供方法学支撑。

科研经历

博士课题 | **基于电子鼻数据的端到端神经网络对母乳风味的识别、预测及模拟研究** | 2023.09 至今

风味分类模型：基于电子鼻电阻响应数据，实现母乳、多种哺乳动物乳与婴配粉的高精度分类。

关键化合物预测模型：构建集成学习框架，由电子鼻响应特征直接预测乳制品中的关键风味活性化合物。

影响因子权重模型：定量解析母体情绪、膳食结构及乳汁脂肪酸/氨基酸组成对母乳风味的贡献权重。

风味模拟优化模型：通过优化婴配粉中脂肪酸与氨基酸的种类及配比，实现配方奶粉风味向母乳风味的定向逼近。

硕士课题 | **牛油渣水解—美拉德反应产生牛肉风味化合物的神经网络预测研究** | 2020.09 – 2023.06

风味预测模型：测定挥发性成分并量化风味物质间的相互作用，实现对样品整体风味轮廓的准确预测。

曲线拟合模型：构建美拉德反应产物中挥发性成分 ROAV 随酶解时间变化的动态拟合曲线。

配方优化模型：从 17 种氨基酸中筛选确定最优添加配比，在最大化风味品质的同时兼顾生产成本。

论文、专利与学术会议

第一作者 SCI 论文（已发表）

Comparative analysis and aroma identification of key flavor components in breast milk, other mammalian milk, and infant formula | *Food Chemistry* | 中科院一区 Top，IF 10.4

Differentiating breast milk and animal milk flavors and investigating influencing factors using electronic nose and machine learning | *Food Bioscience* | 中科院一区，IF 6.2

Prediction of flavor of Maillard reaction product of beef tallow residue based on artificial neural network | *Food Chemistry-X* | 中科院一区，IF 9.2

Process modeling and simulation of key volatile compounds of Maillard reaction products derived from beef tallow residue hydrolysate based on proxy model | *Foods* | 中科院一区，IF 6.0

Quantitative analysis of fatty acids and amino acids as precursors to human milk flavor through importance ranking modeling | Food Bioscience | 中科院一区，IF 6.2

第一作者 SCI 论文（审稿中）

An ensemble learning framework for predicting flavor-active compounds in breast milk based on electronic nose analysis

Artificial neural networks in food flavor analysis: A review

A novel method for simultaneous optimization of multiple processes

中文期刊论文

牛乳与母乳的脂肪酸组成及关键香气化合物的对比分析 ｜ 第二作者 ｜ *中国油脂*，2025

发明专利（已授权）

一种牛油渣美拉德反应产物的牛肉风味改善方法 ｜ 第二发明人 ｜ CN114868897B ｜ 2023.10 授权

一种脂肪酶位置选择性变化规律的评价方法及装置 ｜ 第二发明人 ｜ CN202310612528.3 ｜ 2026.06 授权

发明专利（已公开）

一种基于曲线预测模型的牛油美拉德反应产物挥发性成分预测方法 ｜ 申请号 CN202210864905.8

国际学术会议

第 23 届国际食品科学技术联盟世界大会（IUFoST 2026），中国香港，2026 年 10 月 —— 英文口头报告 2 项，均已录用。

专业技能

感官评价与消费者研究

QDA 定量描述分析、三点检验、偏好测试等感官方法学；评价小组的组建、筛选、培训与一致性校准；感官实验设计与控制性测试环境管理；感官与消费者数据的采集、整理与统计分析。

统计分析与机器学习建模

Python、MATLAB、SQL；方差分析（ANOVA）、主成分分析（PCA）等多元统计方法；神经网络、支持向量机、随机森林、XGBoost、高斯过程回归、K-means 聚类；网格搜索、模型集成与正则化调优；粒子群优化与贝叶斯优化用于工艺与配方优化。

软件开发与流程自动化

Visual Studio 独立开发环境；Python 自动化脚本与浏览器端工具开发；ECharts 数据可视化；MS Office 全套，具备高质量技术文档与汇报材料制作能力。

仪器分析与风味剖析

电子鼻（E-nose）、电子舌（E-tongue）、气相色谱（GC）、气相色谱-嗅闻联用（GC-O）、气相色谱-质谱联用（GC-MS）、全二维气相色谱（GC×GC-MS）；顶空/固相微萃取（HS-SPME）前处理；关键香气化合物的 OAV / ROAV 表征；脂肪酸与氨基酸定量测定。

语言能力

中文（母语）；英语（工作语言）：新加坡国立大学全英文科研环境（2025.09 至今），英文论文写作与国际会议口头报告经验，CET-6。

Jingwei Cui

Ph.D. Candidate in Food Science | Sensory & Consumer Science × Data Science | Flavor Research

Key Strengths

Sensory & consumer research. Strong command of sensory methodology and experimental design, including QDA, triangle tests and preference tests. Experienced in recruiting, screening, training and calibrating sensory panels. Led industry-commissioned sensory evaluation and flavor screening projects, and familiar with data collection standards and quality control in human subject research. Also model individual perceptual differences through variance decomposition and derive consumer preference segments.

Perceptual data science and AI modeling. Proficient in Python, MATLAB and SQL. Independently build, tune and validate neural network, ensemble learning, random forest, XGBoost and Gaussian process regression models in perception research, and apply ANOVA, PCA and other multivariate methods to turn high-dimensional perceptual data into interpretable, decision-ready conclusions.

Multi-source data integration and end-to-end automation. Fuse instrumental sensor data (electronic nose, electronic tongue, GC-MS), human sensory panel data and consumer preference data in a single modeling framework. Sole developer of an automated GC-MS data-processing system (3,000+ samples) and of automated electronic-nose data extraction code (5,000+ samples), automating the full path from raw output to model-ready input. Led the design and governance of a structured analytical database.

Flavor chemistry and instrumental analysis. Independently operate GC-MS, GC-O, GC×GC-MS, electronic nose and electronic tongue. Familiar with identification and quantification of key aroma compounds and with OAV / ROAV characterization, linking instrumental results to sensory conclusions.

Research output and scientific communication. 8 first-author SCI papers (5 published, 3 under review), 2 granted invention patents, and 2 English oral presentations accepted at the IUFoST 2026 World Congress. Able to present findings in English to technical and non-technical audiences as well as to senior leaders.

Education

2023.09 – 2027.06 (expected) Jiangnan University | Ph.D., Food Science and Engineering (Oils and Plant Proteins)
Supervisor: Prof. Xingguo Wang
└ 2025.09 – 2027.06 National University of Singapore | Joint Ph.D. Program, Food Science and Technology (Supervisor: Prof. Shaoquan Liu)

2020.09 – 2023.06 Jiangnan University | M.Sc., Food Science and Engineering (Oils and Plant Proteins)
Supervisor: Prof. Qingzhe Jin | Thesis received dual distinction in blind review

2016.09 – 2020.06 Northeast Agricultural University | B.Sc., Food Science and Engineering

Software Development

Flavor AI Toolkit (personal website) | Sole Developer | redamancy564.github.io
Independently designed and built 8 browser-based AI tools for flavor and perception research, spanning molecules, formulation, instrument data interpretation and consumer perceptual differences.
Core capabilities: predicting odor and taste descriptors from molecular structure; an odor-space map of 5,000 aroma molecules; ingredient and substitute search driven by a flavor brief, individual differences and consumer segmentation; automated GC-MS peak-table parsing with OAV / ROAV ranking of key aroma compounds.
All computation runs locally in the browser with no raw data uploaded, balancing usability with data confidentiality.

Automated GC-MS Data Processing System | Sole Developer
Independently designed, built and maintained an end-to-end software solution automating extraction and analysis of GC-MS datasets following manual peak integration.
Core capabilities: automatic retention-time matching, intelligent aggregation of adjacent peaks, and rule-based selection of flavor compounds when multiple candidates match a single peak.
Impact: reduced secondary data processing from hours to minutes across 3,000+ samples and eliminated manual transcription error; now in routine use for research data processing.

Electronic-Nose Data Processing Code | Sole Developer
Automatically extracts data from raw txt files and performs baseline correction and resistance-value computation, replacing manual per-sample processing.
Processed 5,000+ electronic-nose samples, providing standardized data input for downstream modeling.

Industry R&D Projects

A restaurant chain | Sensory Evaluation Lead 2024.11 – 2024.12
Flavor screening and sensory evaluation
Designed and executed the sensory evaluation protocol, carrying out systematic flavor screening across multiple prototype samples.
Cross-validated instrumental flavor profiling with sensory evaluation to build an objective, reproducible assessment workflow.

A beverage chain | **Project Lead** 2025.01 – 2025.07

Aroma profiling and flavor evaluation of coffee beans

Applied GC-O-MS and ROAV to characterize coffee-bean aroma and established a systematic flavor-profiling and evaluation method.

Built an actionable flavor-screening and evaluation workflow to support product flavor development.

A multinational dairy / infant-nutrition company | **Core Contributor** 2023.09 – 2024.12

Compositional analysis and lipid database construction

Established a standardized sample-preparation and GC quantification workflow for high-throughput fatty-acid analysis of large sample batches.

Led the design and curation of a lipid database, building a structured and reusable analytical data-management system.

A convenience-food company | **Core Contributor** 2023.12 – 2024.04

Quality control and oxidative stability of frying oil

Handled oil quality control and oxidative-stability evaluation, measuring acid value, peroxide value and oxidative stability index among other indicators.

Built a kinetic method for characterizing oil oxidative degradation, providing methodological support for shelf-life assessment.

Research Experience

Ph.D. Thesis | Recognition, prediction and simulation of human milk flavor using end-to-end neural networks based on electronic nose data | 2023.09 – present

Classification model: high-accuracy discrimination of human milk, multiple mammalian milks and infant formula from electronic nose resistance data.

Prediction model: ensemble learning framework predicting key flavor-active compounds directly from electronic nose response features.

Weight model: quantified the contribution of maternal mood, diet, and milk fatty acid / amino acid composition to human milk flavor.

Optimization model: optimized fatty acid and amino acid types and ratios in infant formula to steer its flavor toward that of human milk.

Master's Thesis | Neural network prediction of beef flavor compounds from hydrolysis–Maillard reaction of beef tallow residue | 2020.09 – 2023.06

Flavor prediction model accounting for interactions between volatile compounds to predict overall sample flavor.

Curve-fitting model for ROAV of volatile compounds as a function of enzymatic hydrolysis time.

Additive optimization model selecting the optimal ratio among 17 amino acids, balancing flavor quality against production cost.

Publications, Patents & Conferences

First-author SCI publications (published)

Comparative analysis and aroma identification of key flavor components in breast milk, other mammalian milk, and infant formula | *Food Chemistry* | JCR Q1, IF 10.4

Differentiating breast milk and animal milk flavors and investigating influencing factors using electronic nose and machine learning | *Food Bioscience* | JCR Q1, IF 6.2

Prediction of flavor of Maillard reaction product of beef tallow residue based on artificial neural network | *Food Chemistry-X* | JCR Q1, IF 9.2

Process modeling and simulation of key volatile compounds of Maillard reaction products derived from beef tallow residue hydrolysate based on proxy model | *Foods* | JCR Q1, IF 6.0

Quantitative analysis of fatty acids and amino acids as precursors to human milk flavor through importance ranking modeling | Food Bioscience | JCR Q1, IF 6.2

First-author SCI manuscripts (under review)

An ensemble learning framework for predicting flavor-active compounds in breast milk based on electronic nose analysis

Artificial neural networks in food flavor analysis: A review

A novel method for simultaneous optimization of multiple processes

Chinese-language journal publication

Comparative analysis of fatty acid composition and key aroma compounds in bovine milk and human milk | Second author | *China Oils and Fats*, 2025

Invention patents (granted)

A method for improving the beef flavor of Maillard reaction products derived from beef tallow residue | Second inventor | CN114868897B | Granted Oct 2023

A method and device for evaluating the positional selectivity of lipases | Second inventor | CN202310612528.3 | Granted Jun 2026

Invention patents (published)

A method for predicting volatile components in beef tallow Maillard reaction products based on a curve prediction model | App. No. CN202210864905.8

International conferences

23rd IUFoST World Congress of Food Science and Technology, Hong Kong, October 2026 — two oral presentations accepted (first author).

Technical Skills

Sensory evaluation & consumer research

QDA, triangle tests, preference tests and other sensory methodologies; panel recruitment, screening, training and calibration; sensory experimental design and controlled testing environments; collection, curation and statistical analysis of sensory and consumer data.

Statistical analysis & machine learning

Python, MATLAB, SQL; ANOVA, PCA and other multivariate methods; neural networks, SVM, random forest, XGBoost, Gaussian process regression, K-means clustering; grid search, ensembling and regularization; particle swarm and Bayesian optimization.

Software development & workflow automation

Visual Studio; Python automation scripts and browser-based tool development; ECharts; full MS Office suite with strong technical documentation and presentation skills.

Instrumental analysis & flavor profiling

Electronic nose, electronic tongue, GC, GC-O, GC-MS, GC×GC-MS; HS-SPME sample preparation; OAV / ROAV characterization of key aroma compounds; fatty acid and amino acid quantification.

Languages

Chinese (native). English (working proficiency): full-English research environment at the National University of Singapore since 2025.09, with English publication and international conference presentation experience; CET-6.